Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: google/vit-base-patch16-224

Attack Type: FGSM
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
100.0%	61.97%	44.32%	15.50s
			Accuracy drop: 17.65%

Threat Level Assessment

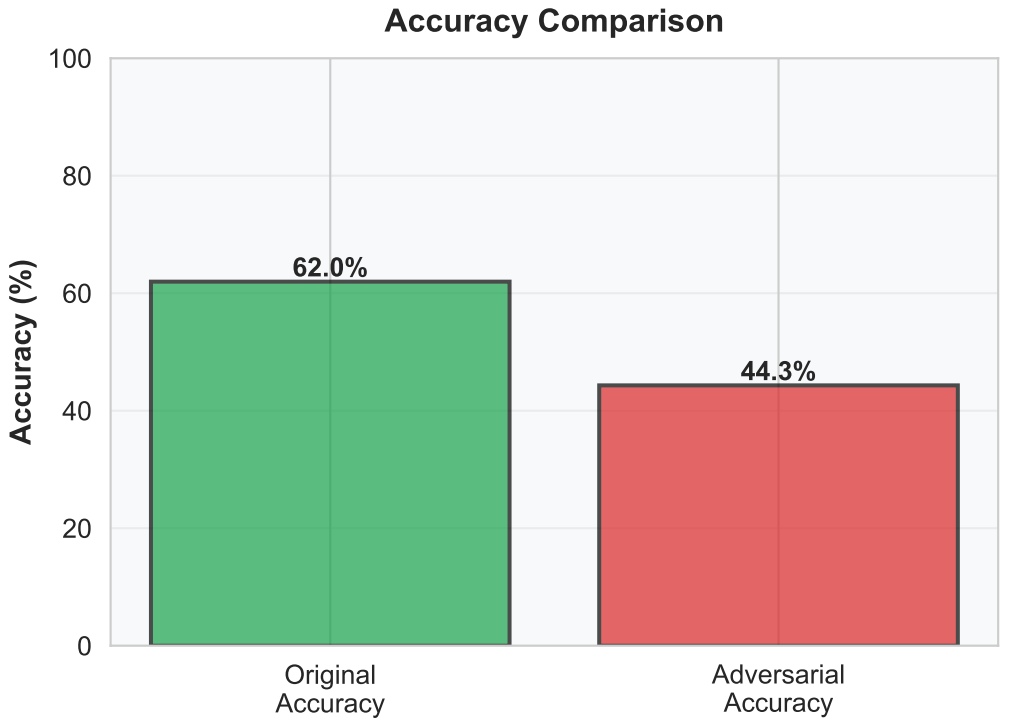
HIGH RISK

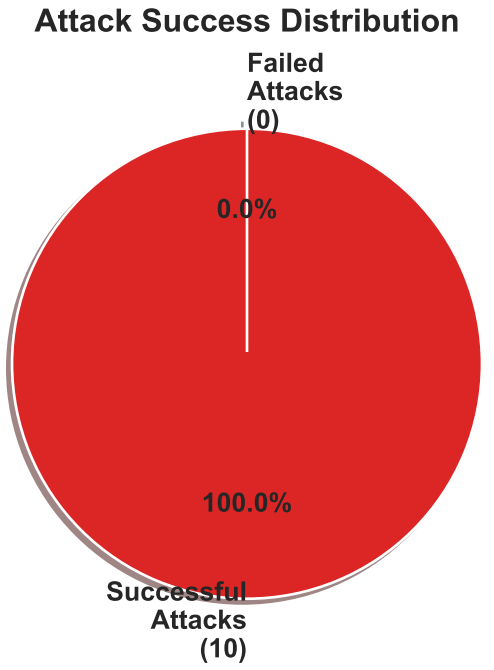
The FGSM attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 61.97% to 44.32%, indicating the current robustness posture is high risk.

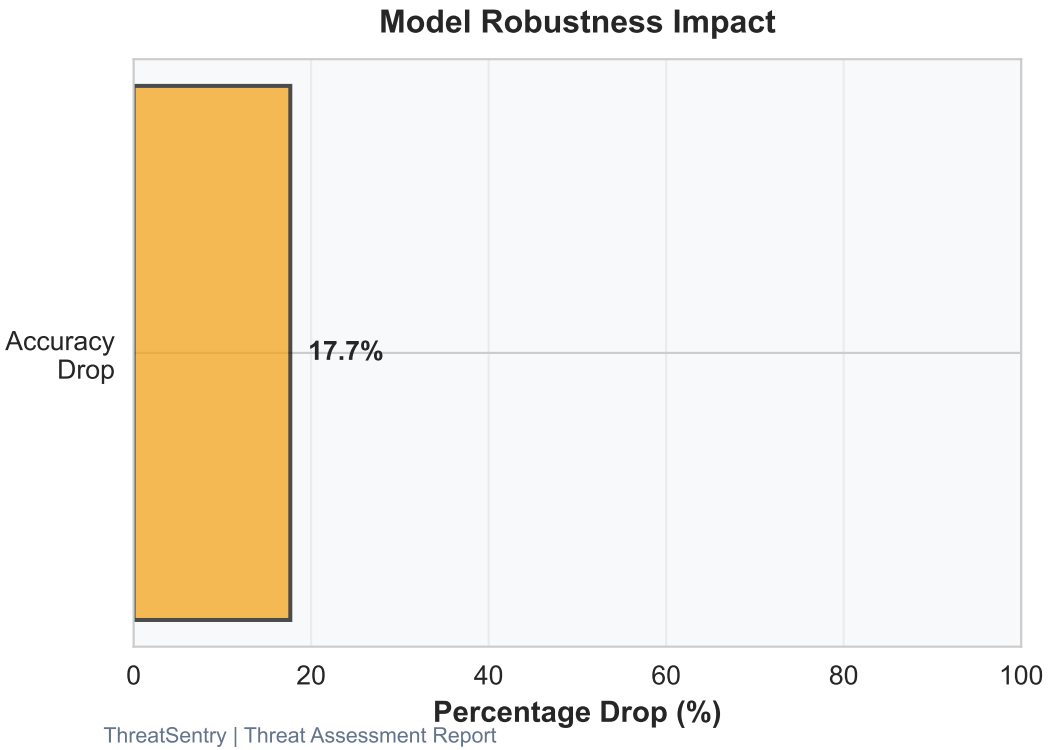
Assessment Narrative

Successfully executed FGSM attack on model google/vit-base-patch16-224 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 61.97%, Average adversarial accuracy: 44.32%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis







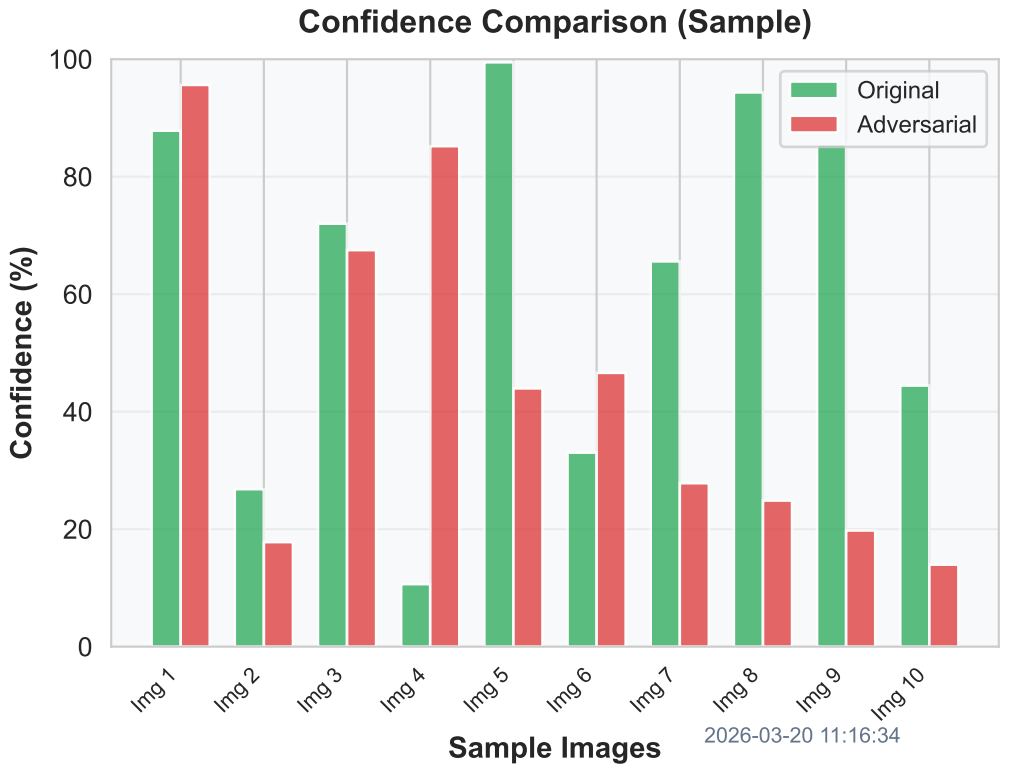
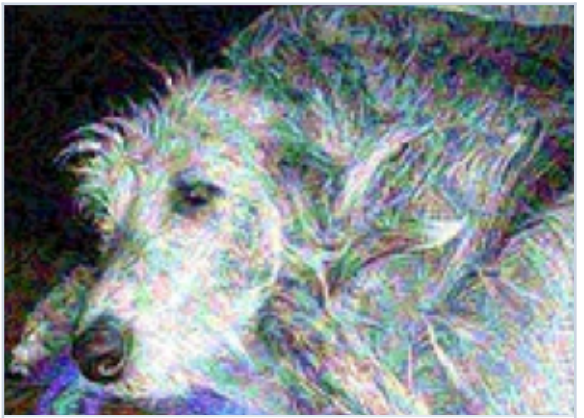


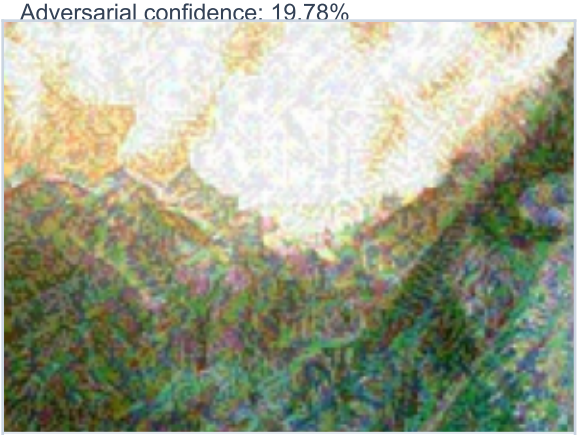
Image Evidence and Predictions

Images 1-4 of 10

Attack Successful
Original image: dugong.JPEG
Adversarial image: dugong.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: dugong, Dugong dugon
Original confidence: 85.51%
After attack predicted as: wool, woolen, woollen
Adversarial confidence: 19.78%



Attack Successful
Original image: Irish_wolfhound.JPEG
Adversarial image: Irish_wolfhound.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Scottish deerhound, deerhound
Original confidence: 94.35%
After attack predicted as: kuvasz
Adversarial confidence: 24.88%



Attack Successful
Original image: alp.JPEG
Adversarial image: alp.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: weasel
Original confidence: 10.65%
After attack predicted as: stole
Adversarial confidence: 85.19%

Attack Successful
Original image: wig.JPEG
Adversarial image: wig.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: maillot
Original confidence: 33.04%
After attack predicted as: poncho
Adversarial confidence: 46.61%

Image Evidence and Predictions

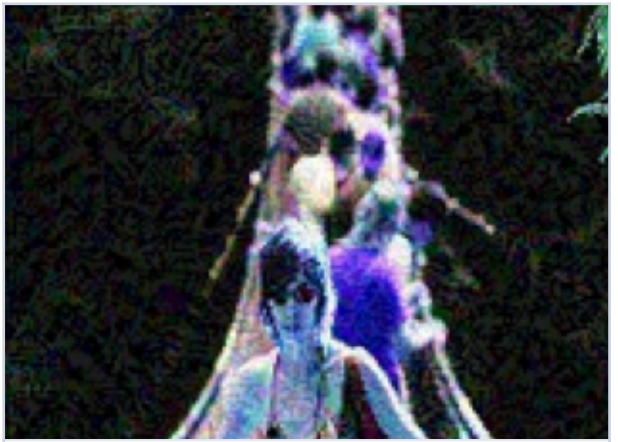
Attack Successful
Original image: cellular_telephone.JPEG
Adversarial image: cellular_telephone.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: hand-held computer, hand-held microcomputer
Original confidence: 87.83%
After attack predicted as: space heater

Attack Successful
Original image: paintbrush.JPEG
Adversarial image: paintbrush.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: milk can
Original confidence: 26.81%
After attack predicted as: schipperke
Adversarial confidence: 17.80%

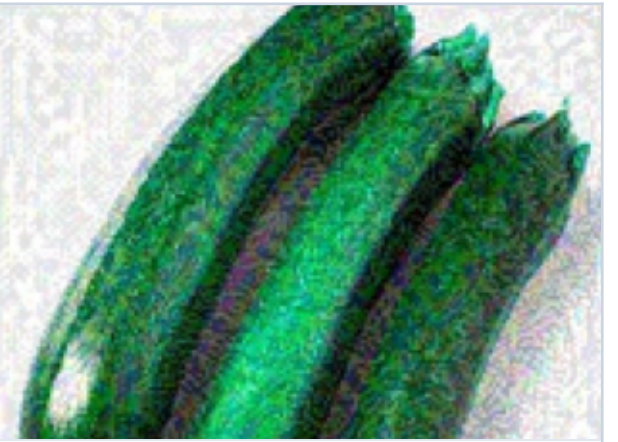
Attack Successful
Original image: black-and-tan_coonhound.JPEG
Adversarial image: black-and-tan_coonhound.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: muzzle
Original confidence: 44.45%
After attack predicted as: brown bear, bruin, Ursus arctos
Adversarial confidence: 13.97%

Attack Successful
Original image: tricycle.JPEG
Adversarial image: tricycle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: bicycle-built-for-two, tandem bicycle, tandem
Original confidence: 65.59%
After attack predicted as: guillotine
Adversarial confidence: 27.83%

Image Evidence and Predictions



Attack Successful
Original image: suspension_bridge.JPEG
Adversarial image: suspension_bridge.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: suspension bridge
Original confidence: 72.00%
After attack predicted as: maypole
Adversarial confidence: 67.50%



Attack Successful
Original image: zucchini.JPEG
Adversarial image: zucchini.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: zucchini, courgette
Original confidence: 99.47%
After attack predicted as: banana
Adversarial confidence: 43.97%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model google/vit-base-patch16-224 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 61.97%, Average adversarial accuracy: 44.32%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques